

Operating Systems Lab Assignment 4 –

Task 1: Batch Processing Simulation

Write a Python script to execute multiple .py files sequentially, mimicking batch processing.

Sol. -

```
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]
# nano script1.py

(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]
# nano script2.py

(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]
# nano script3.py
```

```
root@LAPTOP-2SJNMAE1: /hc  ×  +  v
GNU nano 8.1 script1.py *
print("This is script 1")
|
```

```
root@LAPTOP-2SJNMAE1: /hc  ×  +  v
GNU nano 8.1 script2.py *
print("This is script 2")
|
```

```
root@LAPTOP-2SJNMAE1: /hc  ×  +  v
GNU nano 8.1 script3.py *
print("This is script 3")
|
```

```
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]  
# python3 task1.py  
Running script1.py...  
This is script 1  
Running script2.py...  
This is script 2  
Running script3.py...  
This is script 3
```

Task 2: System Startup and Logging

Simulate system startup using Python by creating multiple processes and logging their start and end into a log file.

```
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]  
# nano task2.py
```

```
root@LAPTOP-2SJNMAE1: /hc × + v  
GNU nano 8.1 task2.py *  
import multiprocessing, logging, time  
  
logging.basicConfig(filename='system_log.txt', level=logging.INFO,  
                    format='%(asctime)s - %(processName)s - %(message)s')  
  
def task(n): logging.info(f"{n} started"); time.sleep(2); logging.info(f"{n} ended")  
if __name__ == "__main__":  
    print("System Booting...")  
    p1=p2=None  
    p1=multiprocessing.Process(target=task, args=("Process-1",))  
    p2=multiprocessing.Process(target=task, args=("Process-2",))  
    p1.start(); p2.start(); p1.join(); p2.join()  
    print("System Shutdown.")
```

```
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]  
# python3 task2.py  
System Booting...  
System Shutdown.
```

Task 3: System Calls and IPC (Python - fork, exec, pipe)

Use system calls (fork(), exec(), wait()) and implement basic Inter-Process Communication using pipes in C or Python.

Sol.-

```
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]  
# nano task3.py
```

```
root@LAPTOP-2SJNMAE1: /hc  ×  +  ▾  
GNU nano 8.1 task3.py *  
import os  
r,w=os.pipe()  
pid=os.fork()  
if pid>0:  
    os.close(r);os.write(w,b"Hello from Parent");os.close(w);os.wait()  
else:  
    os.close(w);msg=os.read(r,1024);print("Child got:",msg.decode());os.close(r)  
|
```

```
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]  
# python3 task3.py  
Child got: Hello from Parent
```

Task 4: VM Detection and Shell Interaction

Create a shell script to print system details and a Python script to detect if the system is running inside a virtual machine.

Sol.-

```
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]  
# nano task4.sh
```

```
root@LAPTOP-2SJNMAE1: /hc  ×  +  v  
GNU nano 8.1 task4.sh *  
#!/bin/bash  
echo "Kernel:"; uname -r  
echo "User:"; whoami  
echo "Hardware:"; lscpu | grep 'Virtualization'
```

```
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]  
# bash task4.sh  
Kernel:  
6.6.87.2-microsoft-standard-WSL2  
User:  
root  
Hardware:  
Virtualization: VT-x  
Virtualization type: full
```

```
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]  
# nano task4.py
```

```
root@LAPTOP-2SJNMAE1: /hc × + ▾
GNU nano 8.1 task4.py *
import platform, subprocess

print("System:", platform.system(), platform.release())
out=subprocess.getoutput("systemd-detect-virt")

print("VM Type:" if out!="none" else "Physical Machine:", out)
|
```

```
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]
# python3 task4.py
System: Linux 6.6.87.2-microsoft-standard-WSL2
VM Type: wsl
```

Task 5: CPU Scheduling Algorithms

Implement FCFS, SJF, Round Robin, and Priority Scheduling algorithms in Python to calculate WT and TAT.

Sol.-

```
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]  
# nano task5.py
```

```
root@LAPTOP-2SJNMAE1: /hc  ×  +  ▾  
GNU nano 8.1 task5.py *  
def avg(l):return round(sum(l)/len(l),2)  
  
def fcfs(bt,at):  
    t=0;ct=[]  
    for i in range(len(bt)):t=max(t,at[i])+bt[i];ct+=[t]  
    tat=[ct[i]-at[i]for i in range(len(bt))];wt=[tat[i]-bt[i]for i in range(len(bt))]  
    print("\nFCFS");[print(f"P{i+1}: WT={wt[i]} TAT={tat[i]}")for i in range(len(bt))]  
    print("Avg WT=",avg(wt),"Avg TAT=",avg(tat))  
  
def sjf(bt,at):  
    n=len(bt);done=[0]*n;t=0;ct=[0]*n  
    for _ in range(n):  
        r=[i for i in range(n)if not done[i]and at[i]<=t]  
        if not r:t=min(at[i]for i in range(n)if not done[i]);r=[i for i in range(n)if at[i]<=t and not done[i]]  
        i=min(r,key=lambda x:bt[x]);t+=bt[i];ct[i]=t;done[i]=1  
    tat=[ct[i]-at[i]for i in range(n)];wt=[tat[i]-bt[i]for i in range(n)]  
    print("\nSJF");[print(f"P{i+1}: WT={wt[i]} TAT={tat[i]}")for i in range(n)]  
    print("Avg WT=",avg(wt),"Avg TAT=",avg(tat))
```

```
root@LAPTOP-2SJNMAE1: /hc  ×  +  ▾  
GNU nano 8.1 task5.py *  
def rr(bt,at,q):  
    n=len(bt);r=bt[:];t=0;ct=[0]*n;Q=[]  
    while 1:  
        for i in range(n):  
            if at[i]<=t and r[i]>0 and i not in Q:Q+=[i]  
        if not Q:  
            if all(x==0 for x in r):break  
            t+=1;continue  
        i=Q.pop(0);x=min(q,r[i]);r[i]-=x;t+=x  
        for j in range(n):  
            if at[j]<=t and r[j]>0 and j not in Q:Q+=[j]  
        if r[i]>0:Q+=[i]  
        else:ct[i]=t  
    tat=[ct[i]-at[i]for i in range(n)];wt=[tat[i]-bt[i]for i in range(n)]  
    print("\nRound Robin");[print(f"P{i+1}: WT={wt[i]} TAT={tat[i]}")for i in range(n)]  
    print("Avg WT=",avg(wt),"Avg TAT=",avg(tat))  
  
def priority(bt,at,pr):  
    n=len(bt);done=[0]*n;t=0;ct=[0]*n  
    for _ in range(n):  
        r=[i for i in range(n)if not done[i]and at[i]<=t]  
        if not r:t=min(at[i]for i in range(n)if not done[i]);r=[i for i in range(n)if at[i]<=t and not done[i]]  
        i=min(r,key=lambda x:pr[x]);t+=bt[i];ct[i]=t;done[i]=1  
    tat=[ct[i]-at[i]for i in range(n)];wt=[tat[i]-bt[i]for i in range(n)]  
    print("\nPriority");[print(f"P{i+1}: WT={wt[i]} TAT={tat[i]}")for i in range(n)]  
    print("Avg WT=",avg(wt),"Avg TAT=",avg(tat))
```

```
bt,at,pr,q=[5,4,2,3],[0,1,2,3],[2,1,3,2],2
fcfs(bt,at);sjf(bt,at);rr(bt,at,q);priority(bt,at,pr)
```

```
root@LAPTOP-2SJNMAE1: /hc × + ∨  
(root@LAPTOP-2SJNMAE1)-[/home/os_assignment4]  
# python3 task5.py
```

FCFS

```
P1: WT=0 TAT=5  
P2: WT=4 TAT=8  
P3: WT=7 TAT=9  
P4: WT=8 TAT=11  
Avg WT= 4.75 Avg TAT= 8.25
```

SJF

```
P1: WT=0 TAT=5  
P2: WT=9 TAT=13  
P3: WT=3 TAT=5  
P4: WT=4 TAT=7  
Avg WT= 4.0 Avg TAT= 7.5
```

Round Robin

```
P1: WT=6 TAT=11  
P2: WT=8 TAT=12  
P3: WT=4 TAT=6  
P4: WT=8 TAT=11  
Avg WT= 6.5 Avg TAT= 10.0
```

Priority

```
P1: WT=0 TAT=5  
P2: WT=4 TAT=8  
P3: WT=10 TAT=12  
P4: WT=6 TAT=9  
Avg WT= 5.0 Avg TAT= 8.5
```